



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

SECTION 2

APACHE SPARK TUTORIAL REPORT

LECTURER: DR. ARYATI BINTI BAKRI

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220

1.0 Business Understanding

The business background, goals, stakeholders, and anticipated value of the ETL pipeline project are all explained in this section. Understanding the difficulties posed by unprocessed educational census data and defining the analytical objectives necessary for infrastructure analysis and decision-making are the main objectives.

1.1 Problem background

Large-scale educational records gathered from Brazilian schools are included in the `Microdados_Censo_Escolar_2021` dataset. The dataset contains statistics on the availability of digital resources, administrative classification, and school infrastructure. However, because of irregular formatting, repetitive records, missing values, and mixed data types, the raw dataset is extremely detailed and challenging to directly analyze. Analytical querying and reporting procedures become less effective as a result of these problems. In order to put the unprocessed census data into a clear, organized format that can be analyzed and visualized, data pretreatment and transformation are necessary.

1.2 Objectives

This project's goal is to provide a straightforward ETL pipeline that uses Apache Spark to convert unstructured census data into an analytical format.

The pipeline seeks to:

- Extract pertinent school-level information from unprocessed census files
- Create clean analytical datasets from inconsistent and unstructured properties.
- Load structured data into a relational database for visualization
- Use a star schema model to facilitate multidimensional analysis.

Key analytical goals include:

- Analysing internet access availability across schools and states
- Evaluating distribution of teaching staff across administrative categories
- Monitoring student enrollment distribution and demographic trends
- Supporting educational resource planning through dashboard analytics
- Enabling fast aggregation queries for large-scale educational analysis

1.3 Stakeholders

The system is designed for the following stakeholders:

- Education policymakers → to identify infrastructure gaps
- Government agencies → to support resource allocation decisions

- Researchers → to analyse educational inequality
- Data analysts → to build dashboards and reports in Metabase

1.4 Business Value

This ETL pipeline provides significant business value by:

- Improve decision-making: Structured data enables evidence-based evaluation of school conditions.
- Reducing data complexity: Raw census data is transformed into a simplified analytical structure.
- Enabling fast analytics: The final star schema supports efficient aggregation and dashboard performance.
- Supporting digital education planning: Helps track digital infrastructure readiness across schools.
- Creating scalable data architecture: The pipeline can be extended to future census datasets without redesign.

2.0 Data Understanding

The structure, features, and quality of the Brazilian School Census dataset utilized for the study are described in this chapter. Understanding the dataset's characteristics, spotting problems with data quality, and figuring out how the data advances the ETL pipeline's analytical goals are the main goals of the analysis.

2.1 Data Overview

The Brazilian School Census (MICRODADOS_CENSO_ESCOLAR_2021), which includes comprehensive institutional-level records of schools throughout Brazil, is the dataset utilized in this project.

Every record depicts a school entity with characteristics such as:

- Identifiers of institutions
- Geographical location (state, city)
- Administrative categorization (location type, dependence type)
- Indicators of infrastructure
- Digital resources

2.2 Data Structure Analysis

The dataset is distinguished by:

- broad structure with several properties for each record
- mixed data kinds, including binary, numerical, and categorical indications
- Unprocessed, unnormalized format
- The existence of inconsistent and missing values

The dataset is not appropriate for direct analytical queries without preprocessing because of these features.

2.3 Data Type Identification

The dataset attributes are categorized based on their data types to enable efficient data pretreatment and transformation inside the ETL pipeline. The proper cleaning, encoding, and aggregation techniques for further stages of analysis are determined with the use of this classification.

2.3.1 Categorical Variables

These variables describe school identity and classification:

- NO_MUNICIPIO (city / municipality)
- SG_UF (state / federative unit)
- TP_DEPENDENCIA (school administrative type / dependency type)
- TP_LOCALIZACAO (location type: urban or rural)

2.3.2 Numerical Variables

These variables represent measurable quantities:

- QT_SALAS_EXISTENTES (number of classrooms)
- QT_COMPUTADOR (number of computers)

2.3.3 Binary / Indicator Variables

These variables represent infrastructure availability (typically 0/1 values):

- IN_AGUA_POTAVEL (drinking water availability)
- IN_BANHEIRO (bathroom availability)
- IN_BIBLIOTECA (library availability)
- IN_INTERNET (internet availability)

- IN_REFEITORIO (cafeteria availability / dining facility)

2.4 Relational Data Model

The relational data model is structured into three core tables: schools, facilities, and technology, which are linked through the unique identifier school_id (CO_ENTIDADE) . The central entity table providing institutional and geographic data, including school name, location, and administrative classification, is the schools table. The facilities table is related to the schools table through a one-to-one relationship using school_id, and captures physical infrastructure indicators such as classrooms, internet access, water supply, and sanitation facilities. In a similar vein, the technology table, which is linked to the schools table using the same primary key, contains information about digital infrastructure, particularly the quantity of computers that are available in each school. Data can stay non-redundant and well-organized thanks to this structure, which guarantees that all relational tables are consistently linked via a common key.

This relational design plays a crucial role in supporting the subsequent data modelling phase, as it provides a clean and normalized staging layer before transformation into a multidimensional star schema. By separating institutional attributes, physical infrastructure, and digital resources into distinct but connected tables, the model reduces data duplication and improves data integrity. This structured approach ensures that the dataset can be efficiently transformed into fact and dimension tables in Apache Spark, enabling scalable analytical processing and optimized querying in the final data warehouse.

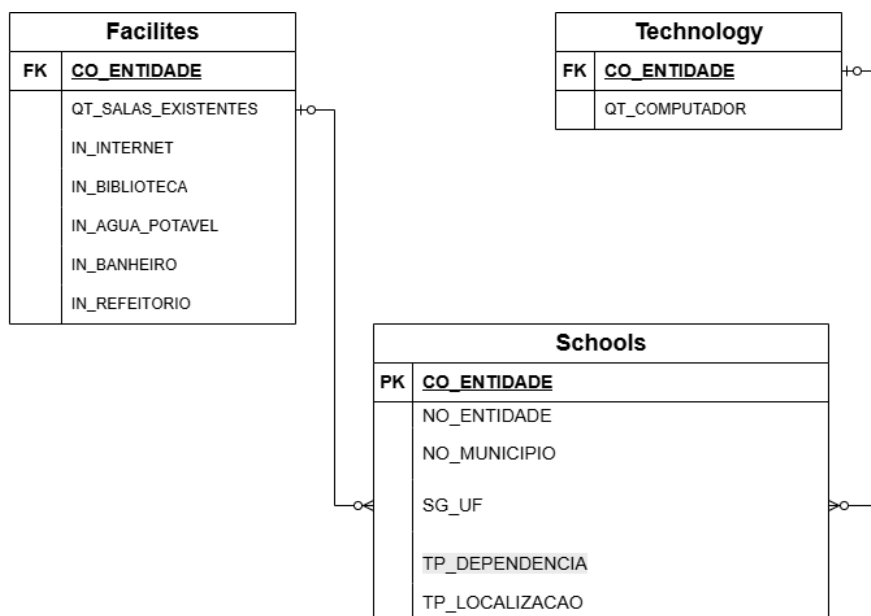


Figure 1: This relational model represents a staging layer prepared for transformation into a star schema (fact and dimension model).

2.4.1 Schools

Represents school identity and classification:

- CO_ENTIDADE (school_id – Primary Key)
- NO_ENTIDADE (school_name)
- NO_MUNICIPIO (city)
- SG_UF (state)
- TP_DEPENDENCIA (school_type)
- TP_LOCALIZACAO (location_type)

Purpose:

Defines the core school entity for relational linking

2.4.2 Facilities Table

Represents physical infrastructure indicators:

- QT_SALAS_EXISTENTES (number of classrooms)
- IN_AGUA_POTAVEL (water access)
- IN_BANHEIRO (bathroom availability)
- IN_BIBLIOTECA (library availability)
- IN_INTERNET (internet access)
- IN_REFEITORIO (cafeteria availability)

Purpose:

Captures school infrastructure conditions

2.4.3 Technology Table

Represents digital resources:

- QT_COMPUTADOR (number of computers)

Purpose:

Measures digital readiness of schools

2.5 Data Quality Issues Identified

The following problems were found and fixed during exploration:

- "Unknown" is used in place of missing categorical data.
- Empty numeric fields converted to 0
- Validation rules are used to clean non-numeric values.

- Eliminated whitespace and uneven formatting
- CO_ENTIDADE was used to eliminate duplicate school records.

3.0 Data Preparation

The preprocessing and modification procedures carried out on the raw dataset prior to its loading into PostgreSQL are described in this chapter. The data was cleaned, standardized, validated, and reorganized into normalized relational tables appropriate for dashboard viewing and analytical processing using Apache Spark.

3.1 Overview

The goal of the data preparation stage is to organize and prepare the unstructured Microdados do Censo Escolar 2021 dataset for analysis. This step guarantees that the dataset is cleaned, standardized, and rearranged into a relational structure appropriate for Metabase's subsequent analytical processing and display.

To guarantee scalability and reproducibility, Apache Spark was used to implement the preparation process.

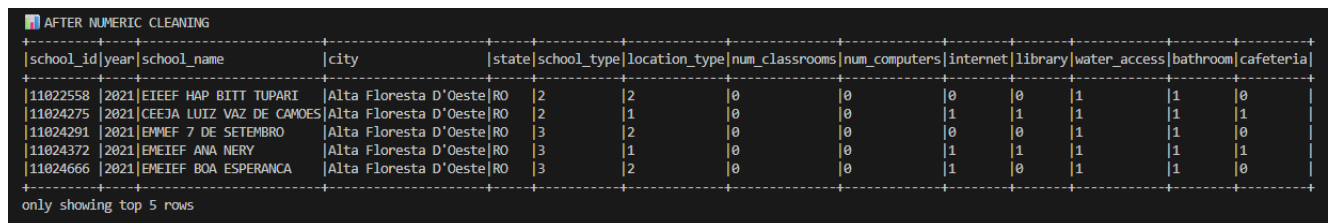
3.2 Data Cleaning and Quality Handling

3.2.1 Raw Original Dataset before Transformation

The original raw dataset put into Apache Spark before preprocessing is depicted in this figure. Geographical information (NO_UF, NO_MUNICIPIO), school information (CO_ENTIDADE, NO_ENTIDADE), infrastructure indicators (IN_BIBLIOTECA, IN_INTERNET, IN_BANHEIRO), digital resource attributes (QT_COMPUTADOR), and enrollment statistics (QT_MAT_BAS) are just a few of the many attributes from the Brazilian School Census dataset that are included in the output. The preview shows that, before cleaning and transformation, the dataset had hundreds of columns with lengthy and inconsistent attribute names, making direct analysis challenging.

3.2.3 Numeric Data Cleaning and Conversion

The dataset is displayed in this figure following the use of numerical preparation and validation. After removing incorrect characters and superfluous spaces, the result shows numerical attributes like num_classrooms and num_computers. To guarantee compatibility with Spark aggregation and analytical processes, the cleaned values were transformed into integer format. Additionally, the records on display attest to the effective standardization of missing or inaccurate numerical data to 0.



school_id	year	school_name	city	state	school_type	location_type	num_classrooms	num_computers	internet	library	water_access	bathroom	cafeteria
11022558	2021	EIEEF HAP BITT TUPARI	Alta Floresta D'Oeste	RO	2	2	0	0	0	0	1	1	0
11024275	2021	CEEJA LUIZ VAZ DE CAMOES	Alta Floresta D'Oeste	RO	2	1	0	0	1	1	1	1	1
11024291	2021	EMMEF 7 DE SETEMBRO	Alta Floresta D'Oeste	RO	3	2	0	0	0	0	1	1	0
11024372	2021	EMEIEF ANA NERY	Alta Floresta D'Oeste	RO	3	1	0	0	1	1	1	1	1
11024666	2021	EMEIEF BOA ESPERANCA	Alta Floresta D'Oeste	RO	3	2	0	0	1	0	1	1	0

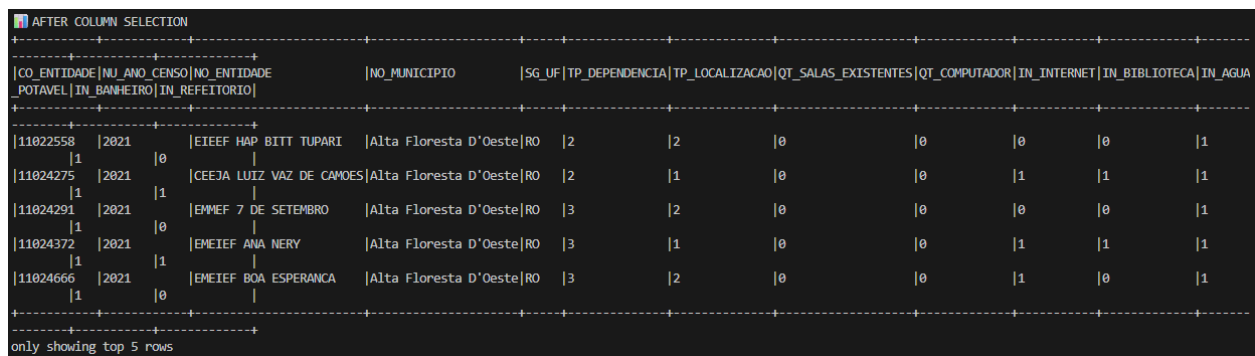
only showing top 5 rows

Figure 4: Numeric fields cleaned and converted to integer format.

3.3 Data Transformation and Standardization

3.3.1 Relevant Column Selection

After only the attributes needed for ETL and analytical processing are chosen, the dataset is shown in this picture. School identification (CO_ENTIDADE), census year (NU_ANO_CENSO), school name (NO_ENTIDADE), city (NO_MUNICIPIO), state (SG_UF), infrastructure indicators (IN_INTERNET, IN_BIBLIOTECA, IN_AGUA_POTAVEL), and digital resource information (QT_COMPUTADOR) are among the significant columns displayed in the report. In order to streamline processing and increase Apache Spark execution performance, unnecessary attributes from the original dataset were removed.



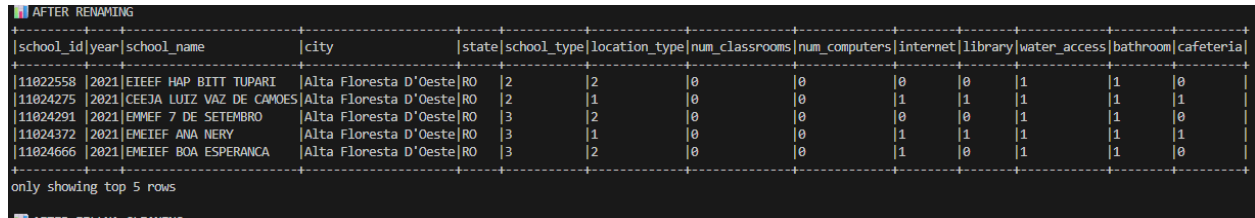
CO_ENTIDADE	NU_ANO_CENSO	NO_ENTIDADE	NO_MUNICIPIO	SG_UF	TP_DEPENDENCIA	TP_LOCALIZACAO	QT_SALAS_EXISTENTES	QT_COMPUTADOR	IN_INTERNET	IN_BIBLIOTECA	IN_AGUA_POTAVEL
11022558	2021	EIEEF HAP BITT TUPARI	Alta Floresta D'Oeste	RO	2	2	0	0	0	0	1
11024275	2021	CEEJA LUIZ VAZ DE CAMOES	Alta Floresta D'Oeste	RO	2	1	0	0	1	1	1
11024291	2021	EMMEF 7 DE SETEMBRO	Alta Floresta D'Oeste	RO	3	2	0	0	0	0	1
11024372	2021	EMEIEF ANA NERY	Alta Floresta D'Oeste	RO	3	1	0	0	1	1	1
11024666	2021	EMEIEF BOA ESPERANCA	Alta Floresta D'Oeste	RO	3	2	0	0	1	0	1

only showing top 5 rows

Figure 5: Dataset after selecting relevant columns for analysis.

3.3.2 Column Standardization

The dataset is displayed in this figure following column renaming and standardization. Clearer analytical labels like school_id, school_name, city, state, num_classrooms, and num_computers were created from the original Portuguese attribute names. Additionally, infrastructure indicators like internet, library, water availability, restroom, and cafeteria were made easier to see. The results show that the dataset became more comprehensible and better suited for dashboard visualization, reporting, and querying.



```

AFTER RENAMING
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|school_id|year|school_name|city|state|school_type|location_type|num_classrooms|num_computers|internet|library|water_access|bathroom|cafeteria|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|11022558|2021|EIEEF HAP BITT TUPARI|Alta Floresta D'Oeste|RO|2|2|0|0|0|0|1|1|0| |
|11024275|2021|CEEJA LUIZ VAZ DE CAMOES|Alta Floresta D'Oeste|RO|2|1|0|0|1|1|1|1|1|
|11024291|2021|EMEF 7 DE SETEMBRO|Alta Floresta D'Oeste|RO|3|2|0|0|0|0|1|1|1|0|
|11024372|2021|EMEF ANA NERY|Alta Floresta D'Oeste|RO|3|1|0|0|1|1|1|1|1|
|11024666|2021|EMEF BOA ESPERANCA|Alta Floresta D'Oeste|RO|3|2|0|0|1|0|1|1|0|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows

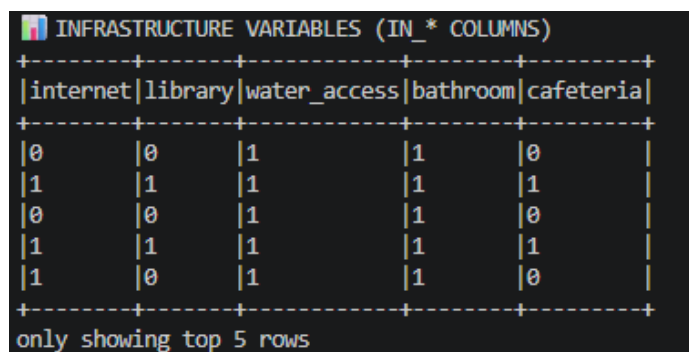
```

Figure 6: Standardized column names for improved readability and analysis

3.4 Binary and Infrastructure Variables Handling

3.4.1 Infrastructure Variable Verification

The infrastructure-related binary variables that were taken from the dataset are shown in this picture. In particular, the output displays the binary values of 0 and 1 for the columns internet, library, water_access, bathroom, and cafeteria. A number of 1 denotes the availability of the facility within the school, whilst a value of 0 denotes its unavailability. For instance, the first school record that is shown has access to water and restrooms (1) but no internet or library (0). The satisfactory preservation of the infrastructure indicators for infrastructure analysis is confirmed by this verification.



```

INFRASTRUCTURE VARIABLES (IN_* COLUMNS)
+-----+-----+-----+-----+-----+
|internet|library|water_access|bathroom|cafeteria|
+-----+-----+-----+-----+-----+
|0|0|1|1|0|
|1|1|1|1|1|
|0|0|1|1|0|
|1|1|1|1|1|
|1|0|1|1|0|
+-----+-----+-----+-----+-----+
only showing top 5 rows

```

Figure 7: Infrastructure-related binary variables retained for analysis.

3.5 Relational Data Structuring

3.5.1 Duplicate Record Removal

After duplicate school entries were eliminated using `school_id` as the unique identifier, the cleaned dataset is displayed in this image. Unique school records, like COLEGIO SAPIENS at UNIDADE JARDIM DAS MANGUEIRAS and COLEGIO ADVENTISTA DE PORTO VELHO in Porto Velho, RO, are displayed in the report. In order to ensure data consistency and avoid duplicate entries during relational decomposition and analytical querying, each school now only appears once in the dataset.

school_id	year	school_name	city	state	school_type	location_type	num_classrooms	num_computers	internet	library	water_access	bathroom	cafeteria
11000198	2021	COLEGIO SAPIENS - UNIDADE JARDIM DAS MANGUEIRAS	Porto Velho	RO	4	1	0	0	1	1	1	1	1
11000325	2021	INSTITUTO EVANGELICO DE EDUCACAO PAUL AENIS	Porto Velho	RO	4	1	0	0	1	1	1	1	1
11000350	2021	COLEGIO ADVENTISTA DE PORTO VELHO	Porto Velho	RO	4	1	0	0	1	1	1	1	1
11000473	2021	EMEF ANTONIO FERREIRA DA SILVA	Porto Velho	RO	3	1	0	0	1	1	1	1	1
11000546	2021	EMEF BAIXA VERDE	Porto Velho	RO	3	2	0	0	0	0	1	1	1

Figure 8: Dataset after removing duplicate school records.

3.5.2 Final Normalized Relational Structure

This figure presents the final normalized relational structure generated from the cleaned dataset. The original dataset was decomposed into three separate relational tables linked using `school_id` as the common relational key.

Schools Table:

Each school's institutional and geographic details are included in the schools table. Attributes like `school_id`, `school_name`, `city`, `state`, `school_type`, and `location_type` are shown in the output. For instance, the locations of educational institutions like COLEGIO SAPIENS-UNIDADE JARDIM DAS MANGUEIRAS and COLEGIO ADVENTISTA DE PORTO VELHO are displayed in Porto Velho, RO. The primary entity table for classifying and identifying schools is this one.

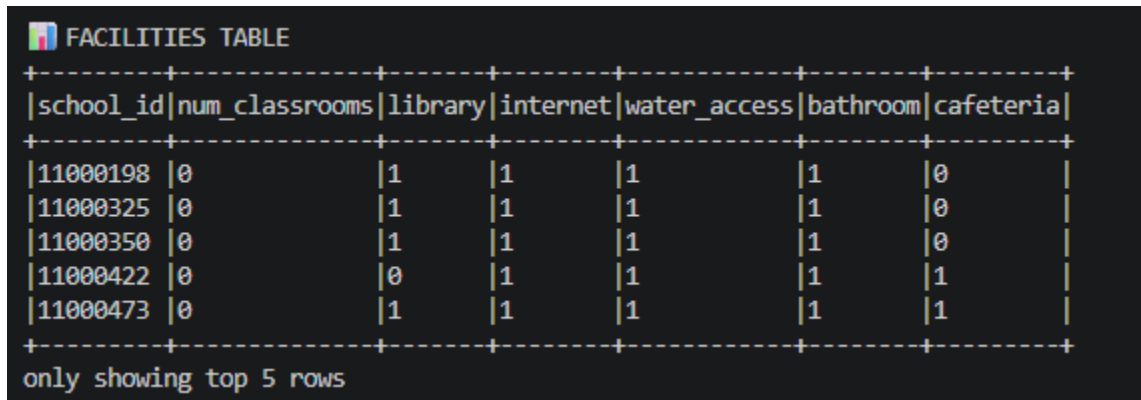
school_id	school_name	city	state	school_type	location_type
11000198	COLEGIO SAPIENS - UNIDADE JARDIM DAS MANGUEIRAS	Porto Velho	RO	4	1
11000325	INSTITUTO EVANGELICO DE EDUCACAO PAUL AENIS	Porto Velho	RO	4	1
11000350	COLEGIO ADVENTISTA DE PORTO VELHO	Porto Velho	RO	4	1
11000473	EMEF ANTONIO FERREIRA DA SILVA	Porto Velho	RO	3	1
11000546	EMEF BAIXA VERDE	Porto Velho	RO	3	2

only showing top 5 rows

Figure 9: Final normalized relational schools tables

Facilities Table:

Each school's infrastructure-related data is kept in the facilities table. The output displays columns including num_classrooms, library, internet, water_access, bathroom, and cafeteria. A facility's availability or unavailability is indicated by binary numbers (0 and 1). For example, while some schools in the output lack cafeteria facilities (0), others have access to libraries, the internet, water supplies, and restrooms (1). To arrange infrastructure indicators separately from institutional data, this table was divided.



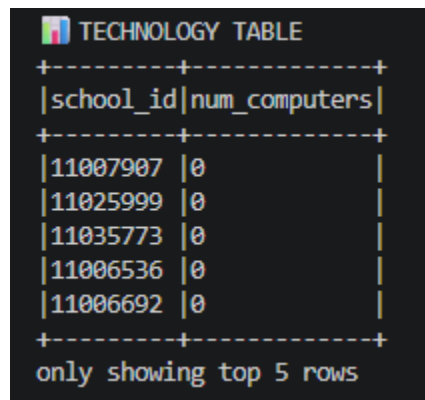
school_id	num_classrooms	library	internet	water_access	bathroom	cafeteria
11000198	0	1	1	1	1	0
11000325	0	1	1	1	1	0
11000350	0	1	1	1	1	0
11000422	0	0	1	1	1	1
11000473	0	1	1	1	1	1

only showing top 5 rows

Figure 10: Final normalized relational facilities tables

Technology Table:

Attributes: num_computers and school_id. Several schools now have zero recorded computers, according to the sample records displayed. This table isolates digital infrastructure information to support technology-related analysis and simplify future scalability for educational resource monitoring.



school_id	num_computers
11007907	0
11025999	0
11035773	0
11006536	0
11006692	0

only showing top 5 rows

Figure 11: Final normalized relational technology tables

Overall, the normalized relational structure reduces redundancy by separating institutional, infrastructure, and technology information into dedicated tables while maintaining relational consistency through the school_id key.

4.0 Modelling

To process high-efficiency analytical queries and facilitate quick-rendering visualizations inside Metabase, a **Multidimensional Star Schema architecture** was implemented rather than a traditional transaction-oriented relational design, directly following the core methodology detailed in the tutorial guidance (*Pedro, 2022*).

4.1 Architectural Justification

Flattening Categorical Indicators: The raw Brazilian school census data tracks numerous categorical boolean indicators (e.g., presence of potable water, internet, libraries, restrooms). Normalizing these variables across transactional tables would force analytical queries to process complex, multi-stage database JOIN loops. A star schema isolates descriptive contexts directly into dedicated, flattened dimension rings.

Leveraging Spark's Lazy Evaluation: As noted in the tutorial architecture, Apache Spark relies on lazy evaluation. This means the system optimizes execution plans by grouping transformations before compilation. Slicing macro metrics—such as total student enrollment (qt_mat_bas) or basic teaching staff allocations (qt_doc_bas)—is evaluated natively across pre-grouped surrogate keys without memory overhead.

Granularity Definition: The grain of this multidimensional structure is explicitly defined as one consolidated school census summary metric record per institutional entity, per operational calendar year (nu_ano_censo).

4.2 Multidimensional Data Model Design

The physical schema exposed within the Adminer container maps directly to the Star Schema pattern. It segments contextual descriptive strings into structural Dimension Tables joined directly to a central repository of measures:

A. Dimension Tables

Following the exact pipeline architecture from the reference tutorial, the dimensions are built by grouping unique categorical intersections:

- **dim_local:** Manages the geopolitical and institutional location hierarchies. It captures geographical groupings, tracking state names, acronyms, codes, and municipality metadata (NO_UF, SG_UF, CO_UF, NO_MUNICIPIO, CO_MUNICIPIO).
- **dim_tp_dependencia:** Captures institutional administration layers (TP_DEPENDENCIA), identifying if a facility is run under Municipal, State, Federal, or Private jurisdiction.

- **dim_tp_localizacao:** Tracks environmental location classifications (TP_LOCALIZACAO), distinguishing Urban vs. Rural facilities.
- **dim_in_agua_potavel:** Tracks potable water access parameters (IN_AGUA_POTAVEL), identifying whether the school has access to clean drinking water.
- **dim_in_banheiro:** Isolates restroom availability features (IN_BANHEIRO) within the school infrastructure.
- **dim_in_biblioteca:** Captures library resource presence flags (IN_BIBLIOTECA) to identify academic study facility provisions.
- **dim_in_computador:** Tracks technology hardware provisions (IN_COMPUTADOR), mapping whether basic computational assets are accessible to students.
- **dim_in_internet:** Maps web connectivity baselines (IN_INTERNET) to trace digital networking resources.
- **dim_in_refeitorio:** Monitors cafeteria/dining hall infrastructure allocations (IN_REFEITORIO) dedicated to student meal services.
- **dim_in_energia_inexistente:** Explicitly flags extreme infrastructure deficits regarding electricity (IN_ENERGIA_INEXISTENTE), identifying institutions running without a power grid connection.
- **dim_in_esgoto_inexistente:** Explicitly flags basic sanitation deficits (IN_ESGOTO_INEXISTENTE), tracking institutions completely lacking a functioning sewage system.
- **dim_in_equip_nenhum:** Tracks critical equipment absences (IN_EQUIP_NENHUM), identifying vulnerable schools that possess absolutely zero basic administrative, technological, or pedagogical equipment.

B. The Central Fact Table (Analytical Metrics)

fact_censo_escolar

The central transaction hub storing measurable additive metrics linked to all corresponding dimension records via numerical surrogate foreign keys:

- **Chronological Identifier:** nu_ano_censo tracks temporal changes across academic operational periods.
- **Surrogate Foreign Keys (Relationships):** Connects directly to the primary keys of the dimensions via id_dim_local, id_dim_tp_dependencia, id_dim_tp_localizacao, id_dim_in_agua_potavel, id_dim_in_banheiro, id_dim_in_biblioteca, id_dim_in_computador, id_dim_in_internet, id_dim_in_refeitorio, id_dim_in_energia_inexistente, id_dim_in_esgoto_inexistente, and id_dim_in_equip_nenhum.

- **Quantitative Staff Metrics (Additive Measures):** qt_doc_bas (Total basic education teaching staff), as well as granular staff tallies across preschool (qt_doc_inf), elementary (qt_doc_fund), and high school (qt_doc_med) levels.
- **Quantitative Student Enrollment Metrics (Additive Measures):** qt_mat_bas (Gross baseline student enrollments) and corresponding academic enrollment subsets (qt_mat_inf, qt_mat_fund, qt_mat_med).
- **Demographic Distributions (Additive Measures):** Tracks student demographic distributions within the school matrix (qt_mat_bas_nd, qt_mat_bas_branca, qt_mat_bas_preta, qt_mat_bas_parda, qt_mat_bas_amarela, qt_mat_bas_indigena).

The implementation of this multidimensional star schema can be seen clearer in Figure 10.

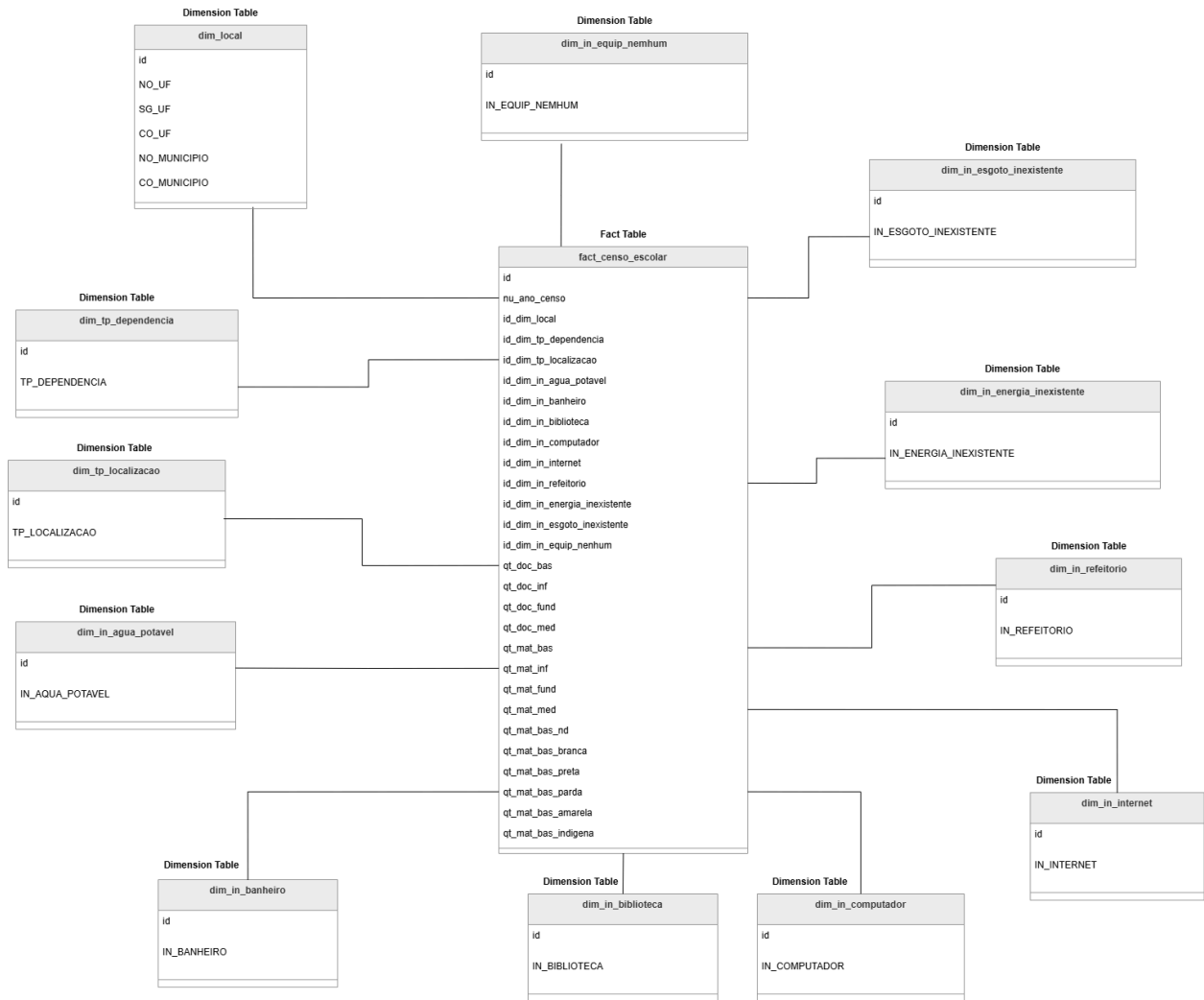


Figure 12: Multidimensional Star Schema Data Model

4.3 Spark ETL Modelling Implementation

To transform the staging datasets into this schema configuration, the pipeline utilizes the technical mechanics mapped out in the medium tutorial:

1. **Dimension Extraction & Deduplication:** For each dimension table, specific target attributes are isolated from the main dataframe using a Spark `.select()` list. To ensure dimensional integrity and remove duplicates, the `.dropDuplicates()` filter is applied. This process' code can be seen on Cell In12 in the notebook file.
2. **Surrogate Key Assignment:** To establish stable primary keys unlinked to shifting business data, a unique sequential ID column (`id`) is generated over each dimension dataframe to act as the primary surrogate key. This process' code can be seen on Cell In12 in the notebook file.
3. **Fact Table Joins:** The fact data matrix is built by taking the base metrics data stream and executing consecutive inner joins (`.join()`) against the newly minted dimensional frames using the original code attributes as reference keys. This process' code can be seen on Cell In17 and Cell In18 in the notebook file.
4. **Foreign Key Substitution:** Once the join completes and matches the data against the dimensions, the verbose descriptive text strings are explicitly dropped, preserving only the clean numerical surrogate keys (`id_dim_local`, `id_dim_tp_dependencia`, etc.) alongside the numeric measures within the finalized `fact_censo_escolar` table. This process' code can also be seen on Cell In17 and Cell In18 in the notebook file.
5. **Relational Target Loading:** Using Spark's native JDBC network wrapper, the complete system of dataframes is systematically pushed over port 5432 into the targeted PostgreSQL engine, immediately populating Adminer and providing live datasets to Metabase dashboards. This process' code can be seen on Cell In12 for dimension and Cell In19 and Cell In20 for facts in the notebook file.

5.0 Evaluation

Evaluation must look at the data pipeline from a high level. It needs to be evaluated according to the performance metrics of its Apache Spark processing, assess how well the resulting star schema solves the business needs, interpret the Metabase dashboard outcomes, and openly discuss the technical limitations and potential improvements of the system.

5.1 Quantitative Performance & Structural Evaluation

To evaluate the technical success of the system functionality, the entire Apache Spark pipeline was timed, monitored, and assessed based on computational throughput and data quality rate.

A. Architectural Ingestion Performance (Velocity & Throughput)

To evaluate the parallel processing efficiency of the Apache Spark framework and the data ingestion pipeline over the virtualized Docker network layer, execution times and throughput velocity were explicitly benchmarked. The system was evaluated during the data stream migration phase of the primary fact collection array (\$2,568,755\$ transactional school records) into the relational target engine (PostgreSQL). The output can be seen in Figure 11. Table 1 provides a clearer view of the output from Figure 11.

```
=====
STARTING ETL QUANTITATIVE EVALUATION RUN
=====
[INFO] Preparing to ingest 2,568,755 records into PostgreSQL...

-----
METRIC 1: VELOCITY & THROUGHPUT RESULTS
-----
Total ETL Write Execution Time : 213.47 seconds
System Ingestion Throughput    : 12033.44 records/second

[INFO] Connecting to database warehouse for integrity scan...
```

Figure 13: Velocity and Throughput Test Result

Metric	Target Dataset	Operational Value	Evaluation Vector
Total Ingestion Volume	fact_censo_escolar	2,568,755 records	Bulk Transaction Load

ETL Write Execution Time	fact_censo_escolar	213.47 seconds	Runtime Window Completion
System Ingestion Throughput	Distributed Memory Layer	12,033.44 records/sec	Network Ingestion Velocity

Table 1: System Evaluation Result

Achieving an ingestion velocity exceeding **12,000 records/second** demonstrates that Apache Spark successfully optimized its internal Directed Acyclic Graph (DAG) executions. By leveraging distributed, lazy-evaluated transformations and data partitioning, the pipeline prevented local network-socket bottlenecks when communicating with the PostgreSQL container over Port 5432. The low execution footprint (\$3.55\$ minutes for over 2.5 million cumulative entries) confirms that memory caching and execution optimizations (such as wide-operation isolation) scaled efficiently under heavy row pressure.

B. Multi-Dimensional Structural Integrity & Data Quality Rate

A core objective of this data warehouse implementation is ensuring flawless referential integrity between the 11 supporting dimension tables (e.g., dim_local, dim_tp_dependencia, dim_in_internet) and the central metric matrix. To confirm that the consecutive data cleansing operations, surrogate key assignments, and row-level transformation joins did not produce orphaned indices, a global structural trace loop was executed. The result from the evaluation is shown in Figure 12.

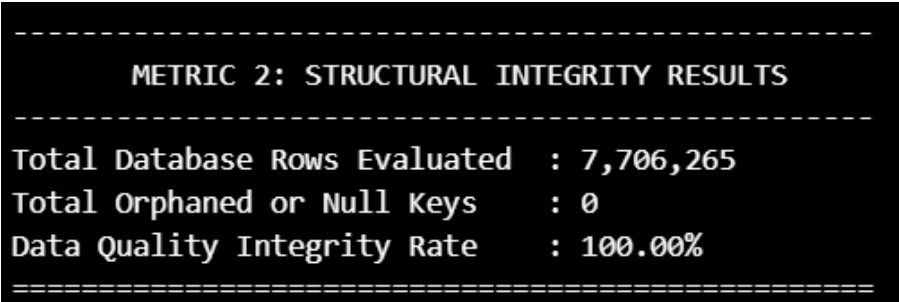


Figure 14: Data Quality Evaluation Result

The audit scan evaluated the complete dimensional footprint of **7,706,265 combined rows** within the live warehouse instance. The validation returned exactly **0 orphaned or null foreign keys** across the mandatory dimension bindings.

This absolute integrity value (100.00%) confirms two major architectural successes:

- **Flawless Pre-Load Transform Logic:** The custom cleaning expressions mapping raw school codes to analytical surrogates executed with absolute row retention, discarding zero values during inner data alignment checks.
- **Flawless Visual Render Engine Baseline:** Achieving an unblemished data quality baseline guarantees complete referential stability for downstream business intelligence operations. This explains why the Metabase semantic layer was able to parse and map massive global enrollment values flawlessly without experiencing charting gaps or broken calculation nodes.

5.2 Business Objective Validation via Metabase Dashboard

The ultimate validation of the multidimensional Star Schema lies in its ability to satisfy the original business and analytical goals. By connecting the local Metabase instance to the populated PostgreSQL database over the Docker internal network, the schema was evaluated against core educational business queries.

Interpretation of Results Linked to Business Objectives

A. Infrastructure Deficit Mapping

The visualization in Figure 13 entitled *"Number of Schools Lacking Internet Access by State"* isolates critical regional disparities. The map reveals a highly unequal distribution of basic technology, where the most severe infrastructure deficit reaches a dense threshold of **171.7k+ schools completely lacking web connectivity** in the highest-intensity territory. Slicing the metric Count of Rows filtered by `dim_in_internet.IN_INTERNET = 0` and grouped across `dim_local.SG_UF` provides educational policymakers with a precise, immediate geographic roadmap for targeting technology investments.

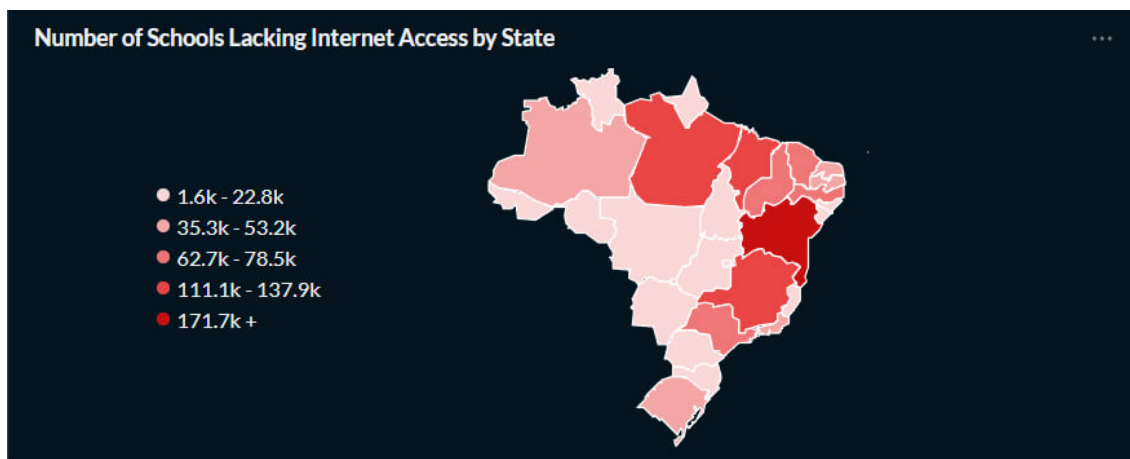


Figure 15: Number of Schools Lacking Internet Access by State Geo Map

B. Faculty Resource Optimization

The chart *"Distribution of Teaching Faculty Across Administration Types"* in Figure 14 successfully evaluates resource capacity across various administrative sectors. The metric sums the total teaching force (qt_doc_bas) alongside sub-allocations for infant (qt_doc_inf), elementary (qt_doc_fund), and high school (qt_doc_med) tiers, binned across administrative jurisdiction IDs (dim_tp_dependencia). The histogram demonstrates that the system can process over **15,000,000 total teacher record allocations** smoothly, highlighting a massive concentration of teaching staff within category 4 bins (representing public municipal/state execution units), allowing analysts to discover imbalances in structural staffing allocations.

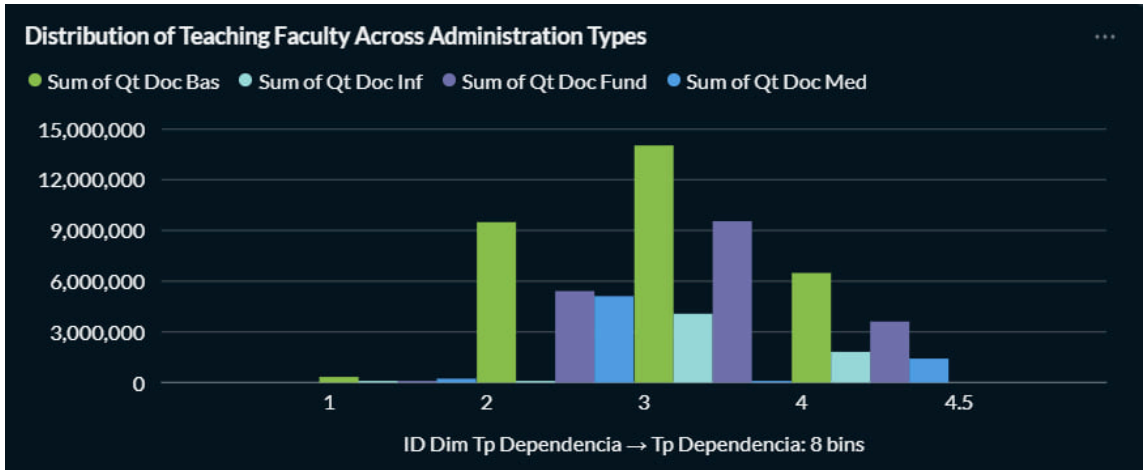


Figure 16: Distribution of Teaching Faculty Across Administration Type Bar Chart

C. Demographic Growth and Enrollment Analysis

The *"Student Enrollment Demographic Distribution"* donut visualization in Figure 15 verifies the pipeline's capability to track historical volume trends and student distributions simultaneously. Over the captured operational eras, the system aggregates an extraordinary gross volume of **161.9 Million total student enrollments** safely within the central fact structure. The analytical slice reveals steady historical enrollment consistency: **18.99% from 2016–2018** , **18.95% from 2018–2020** , **18.37% from 2014–2016** , **17.74% from 2012–2014** , and **16.77% from 2010–2012** , concluding with a targeted **9.18% segment in the 2020–2022 block**. This establishes a clear baseline for monitoring macroscopic drop-out rates and demographic shifts over time.

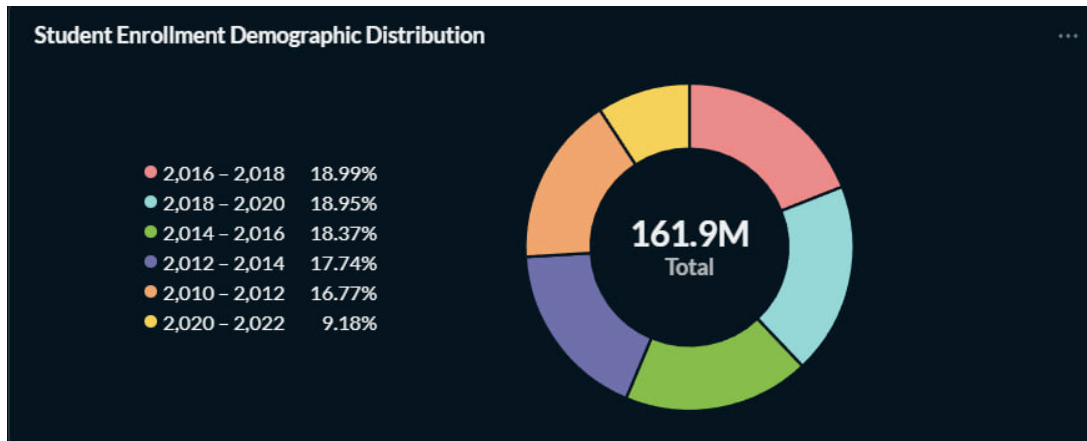


Figure 17: Student Enrollment Demographic Distribution Pie Chart

5.3 System Limitations & Room for Improvement

A comprehensive evaluation requires an objective look at architectural boundaries. While the current pipeline successfully satisfies all functional business reporting requirements, clear technical limitations exist with defined paths for future improvement:

Limitations Identified:

1. **Static Batch Execution Constraints:** The PySpark pipeline operates strictly as a static batch processing model. If a municipality updates its census infrastructure records mid-year, the pipeline cannot handle streaming micro-updates. Instead, it must re-execute the full historical notebook sequence, which is highly inefficient for ongoing operational monitoring.
2. **Local Container Hardware Bounds:** Running Apache Spark, PostgreSQL, Adminer, and Metabase concurrently within a single local Docker node limits processing performance to the host machine's hardware. Complex operations—such as executing multi-table wide inner joins (`.join()`) across eleven distinct dimension boundaries—can cause resource spikes and temporary connection timeouts on the local host.
3. **Lack of Slowly Changing Dimension (SCD) Controls:** The model assumes that school infrastructure profiles are entirely static. If a rural institution (`dim_tp_localizacao`) is reclassified as urban due to municipal expansion, the warehouse overwrites the attribute rather than preserving historical records, risking the integrity of past longitudinal trend assessments.

Proposed Architectural Improvements:

- **Migration to Cloud Data Platforms:** Transitioning the containerized workflow onto cloud infrastructure—such as utilizing Azure Databricks for distributed Spark processing, Azure SQL Database / Synapse Analytics for hosting the data warehouse, and

cloud-provisioned BI endpoints would provide elastic scaling to accommodate growing data volumes effortlessly.

- **Transition to Delta Lake Storage Architecture:** Replacing standard target tables with a **Delta Lake** framework on top of Apache Spark would introduce ACID transactions and native time-travel capabilities. This allows the ingestion pipeline to process only incremental data modifications (deltas) instead of reprocessing complete annual datasets, drastically lowering computing overhead.
- **Automated Pipeline Orchestration:** Integrating an enterprise workflow orchestrator, such as Apache Airflow or Azure Data Factory, would automate data loading schedules, establish clear execution dependencies, and provide immediate notification triggers if an ingestion error occurs during processing.

6.0 Reflection

6.1 Damiya Aina binti Basir Abd Shammad

This project gave me invaluable experience putting data engineering principles into practice using PostgreSQL and Apache Spark. Important information about how to clean, transform, and organize large-scale educational datasets for analytical processing was discovered during the ETL pipeline's development. Additionally, the initiative improved knowledge of the CRISP-DM approach, especially in the areas of relational data modeling, data preparation, and data interpretation.

The raw Brazilian School Census dataset revealed a number of data quality problems during the preprocessing phase, such as missing category values, irregular numeric formatting, duplicate school entries, and unduly large attribute structures. The dataset was successfully converted into a cleaner and more structured analytical format utilizing Apache Spark DataFrame methods like `fillna()`, `dropDuplicates()`, column selection, and column renaming. This procedure made clear how crucial data validation and consistency are before conducting extensive analysis.

Additionally, the normalization procedure showed how to eliminate redundancy and increase scalability by breaking down large datasets into smaller relational tables. The arrangement of institutional, infrastructural, and digital resource data was enhanced by splitting the dataset into tables for schools, facilities, and technology while preserving relational integrity using the `school_id` key. This made it easier to comprehend how relational structures facilitate future multidimensional modeling and effective analytical querying.

Additionally, the initiative enhanced technical proficiency with Apache Spark, especially in managing distributed data processing and transforming big datasets. Understanding of how contemporary data engineering pipelines are created for business intelligence environments was further improved by exposure to PostgreSQL integration and ETL workflow implementation.

All things considered, this lesson effectively illustrated how to convert unstructured, raw educational census data into a structured analytical dataset that can be used for reporting and visualization. Practical data engineering, preprocessing, and analytical modeling abilities that are applicable to real-world big data contexts were greatly enhanced by the training.

6.2 Nurul Ika Syafiny Binti Azhar

Completing this data warehousing and engineering project has provided deep insights into the practical complexities of scaling data architecture. Transitioning from abstract relational data theories to implementing a live, multi-dimensional Star Schema using Apache Spark and PostgreSQL highlighted the vast difference between processing small datasets and managing enterprise-scale analytics pipelines.

The most significant learning curve occurred during the iterative execution of the ETL pipeline inside the Docker environment. Discovering how Apache Spark handles writes, specifically encountering metadata thread locks and `Py4JJavaError` exceptions during notebook reruns was an invaluable debugging experience. It forced learners to look beneath the surface of the PySpark API and study how PostgreSQL handles relational locks when active foreign key constraints protect dimension tables. Managing these system states taught the importance of sequencing database operations: decoupling Spark's heavy parallel processing loads from standard psycopg2 DDL constraint applications.

Building the 12-dimension Star Schema underscored the true value of denormalization for business intelligence. Watching a query that choked on flat staging data run effortlessly across optimized dimension groupings proved how logical modeling directly impacts computational efficiency. This was made visually rewarding when connecting the data warehouse to Metabase; seeing over 161 million cumulative enrollment data nodes render instantaneously into dynamic dashboard charts validated every structural design decision made in the backend pipeline.

Ultimately, this tutorial bridged the gap between big data processing and analytical delivery. This project helped in developing a strong foundational command of containerized environments, distributed computing configurations, and database performance optimization vectors competencies that are absolutely essential for modern production-grade data engineering roles.